

LEC6 - Constellations & Seasons

1/2

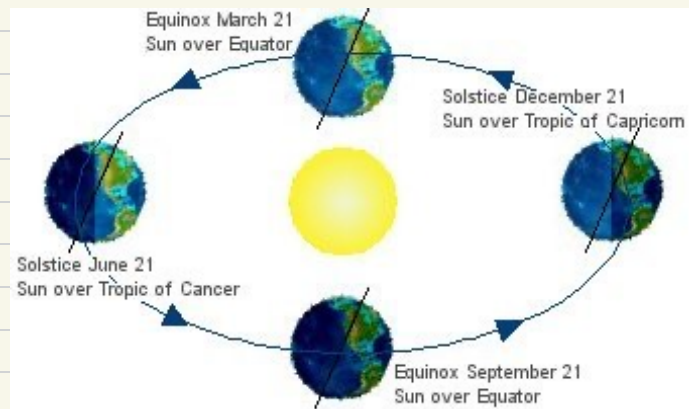
Constellation: A group of stars that appear to form a pattern or picture

Constellations

- Constellations are NOT physically associated
 - ↳ Stars are at different distances
 - ↳ As such, the side view of a constellation would look very different
- We use constellations to navigate to other stars in the sky
 - ↳ Many bright stars have their names indicating their location within a constellation
 - ↳ Alpha, beta, epsilon are greek letters which rank stars in order of brightness
 - ↳ Centaurus, Pictoris, and Eridanus are Western constellations
- We mainly use Roman constellations, but many cultures have their own constellations
- Zodiac constellations are a set of 12 constellations that lie on the **elliptic plane**
 - ↳ Leo is August because the constellation is blocked by the Sun in August

Seasons

- Seasonal variations are **NOT** caused by an elliptical orbit
 - ↳ Earth's orbit is only $\sim 1.5\%$ off of circular
- Seasonal variations occur due to axial tilt (23.4° offset)
 - ↳ Latitudes near equator have more consistent sunlight and therefore less variation
- Northern Hemisphere Summer (June Solstice)
 - ↳ Noon sun hits ground at **steep** angle in **Northern Hemisphere**
 - ↳ Noon sun hits ground at **shallow** angle in **Southern Hemisphere**
- Southern Hemisphere Summer (December Solstice)
 - ↳ Noon sun hits ground at **shallow** angle in **Northern Hemisphere**
 - ↳ Noon sun hits ground at **steep** angle in **Southern Hemisphere**
- When the Sun is high in the sky during the day, the number of daylight hours is greater and the amount of direct sunlight received is greater
 - ↳ contributes to summer
- When the sun is low during the day, the number of daylight hours is less and the amount of direct sunlight received is less
 - ↳ contributes to winter
- Distance from Sun doesn't matter because the variation is small ($\sim 3\%$)
 - ↳ Sun is closest during Northern winter, and farthest during Northern summer
- Planets with more variation in distance have distance play a greater role in seasons (Ex. Pluto)
 - ↳ Distance to Sun varies from 29AU to 49AU
- Equinox means equal length days and nights
- Solstice means shortest day/night
- Seasonal differences would be **more** extreme if axial tilt was greater (=Uranus 97°)
- Seasonal differences would be **less** extreme if axial tilt was even greater (=Venus 177°)
 - ↳ Even though tilt is greater 177° is only 3° from being completely neutral



- Constellations have great cultural significance, but are mainly used for navigation in Astronomy
- Seasons are not caused by varying distance from the Sun, and are instead caused by the Axial tilt of Earth

LEC6 - Constellations & Seasons

2/2

Milankovitch Cycles:
The collective changes
in the earth's movement
VIA eccentricity, axial
tilt, and rotational axis

- Earth's rotational axis is **NOT** fixed

- ↳ takes 26,000 years for earth's North pole to make 1 precession

- Axial tilt varies from 22.5° - 24.5° every roughly 41,000 years

- Eccentricity varies from 3-6% elliptical roughly every 100,000 years

} Known as
Milankovitch Cycles

• The eccentricity, rotational axis, and axial tilt of earth all vary slightly over thousands of years in a cycle